Chandrika Chandra | AI/ML Python Engineer
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Professional Summary

A **highly skilled and results-oriented AI/ML Python Engineer** with over 3 years of experience in developing and deploying full-stack, data-driven solutions. I specialize in designing and implementing **end-to-end machine learning pipelines**, from data acquisition and preprocessing to model deployment and monitoring. My expertise includes leveraging **Python, PySpark, and leading ML frameworks like TensorFlow and scikit-learn** to build predictive and analytical models that solve complex business challenges. I am adept at creating scalable, robust, and maintainable systems using **cloud platforms (AWS), REST APIs, and modern DevOps practices (Docker, Jenkins)**. With a strong background in data engineering and a passion for turning raw data into actionable insights, I consistently deliver solutions that drive strategic decision-making and business growth.

Technical Skills

Languages:	Python, SQL, Bash
ML/AI	scikit-learn, TensorFlow, Keras, XGBoost, PyTorch (basics)
Data Engineering	PySpark, Pandas, NumPy, Spark SQL, ETL
Visualization	Matplotlib, Seaborn, Plotly, Tableau (basic)
Web Development	Flask, Django, REST APIs
Cloud Platforms	AWS (EC2, S3, RDS, IAM, Route 53, VPC)
Databases	MySQL, PostgreSQL, NoSQL
DevOps/Tools	Git, Jenkins, Docker, Jupyter, Jira
Testing	PyTest, PyUnit
OS/IDEs	Linux, Windows, PyCharm, Jupyter Notebook, VS Code

Professional Experience

AI/ML Python Engineer | Emory Healthcare - Atlanta, GA
August 2024 – June 2025

- Engineered and productionized end-to-end **machine learning pipelines** for predictive analytics on massive healthcare datasets using **scikit-learn**, **XGBoost**, and **LightGBM**, achieving a 15% improvement in operational efficiency.
- Developed a real-time, web-based dashboard using **Flask**, **Plotly**, and **Dash** to visualize model predictions, KPIs, and drift metrics, providing business teams with actionable insights.
- Built **Natural Language Processing (NLP)** models using **spaCy**, **NLTK**, and **TF-IDF** to extract insights from physician notes, improving clinical documentation analysis accuracy by 22%.
- Designed and deployed **deep learning models** using **TensorFlow** and **Keras** for medical image classification and sequential time-series prediction.
- Created automated **model retraining pipelines** triggered by data drift or scheduled intervals using **Python**, **Airflow**, and **AWS Lambda**, ensuring model freshness.
- Leveraged **PySpark** and **Pandas** on distributed clusters to perform feature engineering and large-scale ETL transformations across structured and unstructured datasets.
- Developed scalable, containerized **RESTful APIs** with **Flask** and **FastAPI** for serving real-time ML inferences to internal dashboards and client applications.
- Deployed models in production using **AWS SageMaker**, **Docker**, and **EC2**, with auto-scaling and health-check monitoring for high availability.
- Designed custom **model evaluation frameworks** utilizing **ROC-AUC**, **PR curves**, **F1-scores**, and **SHAP** for explainability, ensuring regulatory compliance and ethical AI practices.

- Implemented **hyperparameter optimization** using **GridSearchCV**, **RandomizedSearchCV**, and **Optuna**, improving model performance while minimizing overfitting.
- Monitored deployed models using **Prometheus** and **Grafana**, and configured alerting systems for performance degradation and data quality issues.
- Built **CI/CD pipelines** for ML using **Jenkins**, **GitHub Actions**, and **Docker**, reducing deployment time by 40% and enabling reproducible training runs.
- Managed relational data using **PostgreSQL** on **AWS RDS**, ensuring secure, consistent access to training and prediction data for batch jobs.
- Worked in close coordination with **data scientists**, **DevOps engineers**, and **product managers** to define business objectives, convert them into technical models, and present insights to leadership and non-technical stakeholders.

Python Developer (with ML Tasks) | HCL, Hyderabad, India
February 2021 – July 2023

- Developed and deployed **end-to-end ML solutions** for internal tools, including a **customer churn prediction model and a sentiment analysis engine**, directly impacting product strategy and customer retention.
- Implemented and fine-tuned regression and classification models using **scikit-learn**, visualizing model performance and findings with **Seaborn and Plotly** for a clear and comprehensive understanding of results.
- Created robust and efficient **data processing pipelines with Pandas and Spark SQL** to clean, transform, and aggregate disparate data sources, significantly improving the quality and reliability of model inputs.
- Built and managed a robust **API layer using the Django REST Framework** to expose ML model outputs, enabling seamless consumption by internal and external applications.
- Authored and maintained **unit tests with PyTest** for all critical Python and ML modules, ensuring the reliability, robustness, and maintainability of the codebase.
- **Automated data ingestion workflows** for various data formats (e.g., CSV, JSON) and integrated them into the data pipeline, reducing manual effort and ensuring timely data availability.
- **Designed and executed A/B testing frameworks and statistical analyses** to quantify the business impact of new models and features, providing data-backed evidence for decision-making.
- **Contributed to the development of a data governance framework** by creating Python scripts to validate data schema and ensure data integrity before model training.
- **Leveraged Git for version control and collaborative development** of all code, including feature branches, pull requests, and code reviews, ensuring a high standard of code quality.
- **Worked closely with DevOps teams to containerize ML applications** using Docker, standardizing the environment and simplifying deployment across different stages.
- **Developed comprehensive documentation for ML models and pipelines**, including data dictionaries, model cards, and usage guides, to facilitate knowledge sharing and team collaboration.

Education
Master of Science in Artificial Intelligence
University of North Texas 2025

Bachelor of Engineering in Computer Science & Engineering
Guru Nanak Institute of Technical Campus, India